

State and proof roadmap for Lemmas

Quantitative strict-AT formalization

Snapshot checked on 1 August 2026

1 Executive status

The library target `LatalaMeetsAT` builds successfully with Lean 4.32.1 and mathlib 4.32.1. The directory `Lemmas` contains 26 Lean files. Sixteen files have no local `sorry`; ten files contain 13 occurrences in total. There are no project-specific `axiom` declarations and no `admits` in this directory.

The build is therefore structurally complete but not certified. Lean reports that `SpinGlass.AT.quantitative_strictAT` depends on `sorryAx`, in addition to the usual foundations `propext`, `Classical.choice`, and `Quot.sound`. A file marked “locally complete” below has no `sorry` in that file; it may still depend transitively on an open declaration imported earlier.

Three of the 13 holes are definitions rather than propositions: `scalarOrderParameter`, `gtSemigroupSolution`, and `cavityInterpolatedAverage`. These are foundational interfaces. Filling later theorem holes before fixing these interfaces would hide the main analytic work rather than remove it.

Since the preceding snapshot, the scalar trial identity `scalarTrialValue_eq` has been proved from a reusable Gaussian convolution argument. The proof includes integrability of $\log \cosh$ under Gaussian measures, rescaling of two independent Gaussian variables, variance addition, and the endpoint $q = 0$. The target build completes in 3154 jobs.

2 Import spine

The displayed chain is the principal dependency spine. The cavity portion now has small branches: coefficients depend on definitions, the system additionally uses interpolation, stability depends on coefficients, and absorption joins stability, the system, and fixed-deviation estimates.

```
Replicas → RSPParameters → UniformData → SmartPath → GaussianDerivative →
FreeEnergyDerivative → Scalar/Semigroup → Scalar/LatalaKernel → Scalar/StrictATSign
→ GT/Defs → GT/Semigroup → GT/HalfCascade → GT/Interpolation → GT/Coercivity →
CoupledPressure → FixedDeviation → Cavity/Defs → Cavity/Interpolation → Cavity/
Coefficients → Cavity/System → Cavity/Stability → Absorption → FreeEnergy → Replicon
→ CompactCorollary → MainResult.
```

3 State of every Lean file

File	Local state	What is present or missing
<code>Replicas.lean</code>	locally complete	Finite configurations, positive partition functions, Gibbs-weight bounds and normalization, permutation invariance, overlap bounds, linearity, monotonicity, and the mixed-overlap estimate.
<code>RSPParameters.lean</code>	locally complete	Gaussian RS quantities, the selected fixed point, positivity and range of q , identities for A , and anomalous-versus-replicon comparison.

File	Local state	What is present or missing
<code>UniformData.lean</code>	locally complete	Compact-uniform parameter package and the strict-AT path gap.
<code>SmartPath.lean</code>	locally complete	Covariance kernel, abstract centered Gaussian path, full Hamiltonian with external field, overlap second moment, and path gap. An explicit finite Gaussian realization is not supplied.
<code>GaussianDerivative.lean</code>	open (1)	The explicit covariance operator is present; the reusable finite-Gaussian differentiation theorem remains.
<code>FreeEnergyDerivative.lean</code>	open (1)	Definitions and <code>rsGap_deriv</code> are complete; <code>smartPath_freeEnergy_deriv</code> remains.
<code>Scalar/Semigroup.lean</code>	locally complete	Both branches of <code>scalarPsi</code> and <code>scalarTrialValue_eq</code> are proved, including Gaussian integrability and convolution.
<code>Scalar/LatalaKernel.lean</code>	open (1)	The derivative and sign of <code>latalaH</code> , kernel antitonicity, opposite-monotonicity covariance inequality, and weighted comparison are proved. Only the singular reference-mean identity remains.
<code>Scalar/StrictATSign.lean</code>	open (2)	The scalar diffusion order parameter and all strict sign estimates remain.
<code>GT/Defs.lean</code>	locally complete	Attainable overlaps, constrained free energy, signed matrix path, mass profile, correction, terminal condition, and endpoint identities.
<code>GT/Semigroup.lean</code>	locally complete	Elementary zero, half, and one semigroup definitions; only continuity of the zero operator is proved.
<code>GT/HalfCascade.lean</code>	locally complete	Constant-function half-cascade identity.
<code>GT/Interpolation.lean</code>	open (2)	The two-dimensional recursion and the two-replica Guerra–Talagrand bound remain.
<code>GT/Coercivity.lean</code>	open (1)	The local positive constant and abstract quadratic Taylor-loss lemma are proved; uniform GT quadratic coercivity remains.
<code>CoupledPressure.lean</code>	open (1)	Correct quadratic pressure definitions are present. The uniform sublinear overlap estimate remains; its public alias is complete.
<code>FixedDeviation.lean</code>	open (1)	Exponentially small fixed-deviation tails remain.
<code>Cavity/Defs.lean</code>	locally complete	The observables A, B, C , the identity between A and the overlap second moment, cavity vector, source vector, and 3×3 cavity matrix.
<code>Cavity/Interpolation.lean</code>	open (2)	The elementary third-moment bound is complete. The last-spin interpolation and its second-derivative estimate remain.
<code>Cavity/Coefficients.lean</code>	locally complete	Replica-edge classification, decoupled coefficients, and equality with the displayed cavity matrix.
<code>Cavity/System.lean</code>	conditional	No local hole. The system is an algebraic identity because the remainder is defined as the residual. Its analytic bound is a predicate and is assumed, not constructed.

File	Local state	What is present or missing
<code>Cavity/Stability.lean</code>	locally complete	Explicit matrix inverse with two-sided identities, triangular determinant formula, anomalous and replicon gaps, uniform inverse estimate, and replicon left eigenvector.
<code>Absorption.lean</code>	locally complete	Pointwise cubic splitting, exact cavity-vector norm, domination of B and C by A , and the uniform NA_s bound are proved from the stated cavity-remainder hypothesis.
<code>FreeEnergy.lean</code>	open (1)	RS and finite-volume free energies are defined; the uniform $O(N^{-1})$ error remains.
<code>Replicon.lean</code>	locally complete	Third-moment little- o and replicon susceptibility are proved from the earlier tail, absorption, stability, and assumed cavity-remainder results.
<code>CompactCorollary.lean</code>	locally complete	Compactness plus continuity and strict AT produce <code>UniformATData</code> .
<code>MainResult.lean</code>	conditional	<code>quantitative_strictAT_of_cavity_remainder</code> assembles the three conclusions from an explicit remainder bound; the compatibility wrapper retains the existential hypothesis. Transitive holes yield <code>sorryAx</code> .

4 Difficulty and recommended prover for open files

The rating is for the file in its present repository state, including missing helper lemmas and definitions. “Codex” means it should lead the work when the task requires designing definitions, repairing APIs, adding supporting lemmas, or coordinating changes across files. “Aristotle” means the statement and interfaces are already stable enough for focused Lean proof search. A handoff to Aristotle is useful only after the named prerequisite has been supplied. No open file is rated easy at present.

File	Difficulty	Better lead	Reason and possible handoff
<code>GaussianDerivative.lean</code>	difficult	Codex	A core definition and a finite-Gaussian differentiation API are absent. Hand off only the final finite-sum and integration-by-parts proof after that API is fixed.
<code>FreeEnergyDerivative.lean</code>	difficult	Codex	The proof must instantiate the Gaussian API, derive the covariance cancellation, and normalize replica sums. Aristotle can help with the final algebra after those identities are named.
<code>Scalar/LatalaKernel.lean</code>	difficult	Codex	The remaining theorem combines an endpoint-singular reference density, the substitution $y = 1 - u^2$, and an interchange with the Gaussian integral. The pointwise antitonicity and weighted comparison are already complete.
<code>Scalar/StrictATSign.lean</code>	difficult	Codex	The order parameter itself is missing and requires a PDE or stochastic-process interface. This is definition and architecture work before proof search can begin.

File	Difficulty	Better lead	Reason and possible handoff
<code>GT/Interpolation.lean</code>	difficult	Codex	The two-dimensional recursion, cascade construction, endpoint equations, and derivative sign must be built together. Aristotle can close local endpoint or algebraic sublemmas once extracted.
<code>GT/Coercivity.lean</code>	difficult	Codex	The proof needs new multiplier-derivative, Taylor, compactness, and signed-range lemmas. Its present single theorem is too broad for isolated proof search.
<code>CoupledPressure.lean</code>	difficult	Codex	The argument coordinates GT coercivity, overlap discretization, Gaussian concentration, convexity, Gronwall, and a uniform limit sequence.
<code>FixedDeviation.lean</code>	difficult	Codex	The abstract disorder lacks the explicit coordinate or concentration interface needed for the gradient estimate. The model-level gap must be repaired first.
<code>Cavity/Interpolation.lean</code>	difficult	Codex	The last-spin interpolation is undefined, and the derivative expansion requires systematic replica bookkeeping and uniform estimates. Aristotle can handle bounded coefficient identities after they are separated.
<code>FreeEnergy.lean</code>	moderate	Codex	The source still needs endpoint and closed-interval calculus lemmas. After Codex adds those interfaces, Aristotle can finish nonnegativity and the uniform M/N estimate.

5 The 13 remaining sorry obligations

5.1 Gaussian interpolation and free-energy derivative

1. `GaussianDerivative.lean:quenchedGibbs_deriv_of_covariance_deriv`. Prove that the explicit `replicaCovarianceOperator` is the derivative of the quenched finite-replica expectation. The missing input is smooth dependence of a finite-dimensional centered Gaussian law on its covariance, or an explicit affine Gaussian realization with the required domination bounds.
2. `FreeEnergyDerivative.lean:smartPath_freeEnergy_deriv`. Apply the finite-Gaussian theorem, prove the diagonal cancellation, and rewrite the finite replica sums as `overlapSecondMoment`. The target derivative is $\frac{\beta^2}{4}((1-q)^2 - \mathbb{E}\langle(R_{12}-q)^2\rangle)$.

5.2 Scalar Latala comparison

3. `Scalar/LatalaKernel.lean:latalaF_reference_mean_one`. Prove $\int_0^1 F_\lambda(y)/(2\sqrt{1-y}) dy = 1$. Under $y = 1-u^2$, the weighted inner integral reduces to the antiderivative of $u/(1+(1-u^2)t)^2$. The formal proof must justify the endpoint singularity and the interchange with the Gaussian integral.

4. `Scalar/StrictATSign.lean:scalarOrderParameter`. Construct $g_s(u) = \mathbb{E}[(\partial_x \Psi_s(u, X_u))^2]$ for the local-field diffusion, including the piecewise dynamics at q .
5. `Scalar/StrictATSign.lean:strictAT_sign`. Below q , use the diffusion derivative and conditional Jensen. Above q , apply the proved weighted Latała comparison. Integrate the slope bounds and obtain uniform local constants by compactness.

5.3 Guerra–Talagrand interpolation

6. `GT/Interpolation.lean:gtSemigroupSolution`. Define the finite two-dimensional Parisi recursion for `signedMatrixPath`, `gtMassParameter`, and `gtTerminal`, with its integrability and endpoint properties.
7. `GT/Interpolation.lean:twoReplica_GT_bound`. Build the finite half-mass cascade, identify both endpoints, and prove that the Gaussian interpolation derivative is nonpositive. At zero multiplier, prove factorization into the two scalar RS semigroups and match `gtCorrection`.
8. `GT/Coercivity.lean:gt_quadratic_coercivity`. Differentiate the GT functional in its multiplier, combine `strictAT_sign` with a uniform second-derivative estimate, use compactness away from q , and invoke the proved abstract Taylor-loss lemma near q .

5.4 Coupled pressure and concentration

9. `CoupledPressure.lean:coupledPressure_sublinear`. Split over the $N + 1$ attainable overlaps, apply GT coercivity and Gaussian concentration, then combine the resulting envelope with convexity, `rsGap_deriv`, and Gronwall to obtain a uniform sequence tending to zero.
10. `FixedDeviation.lean:fixedDeviation`. Prove the uniform $O_K(N)$ squared-gradient estimate for the quadratic exponential moment and apply Gaussian concentration. This needs an explicit finite Gaussian coordinate model or a concentration theorem for `RSSmartPathDisorder`.

5.5 Cavity interpolation and free energy

11. `Cavity/Interpolation.lean:cavityInterpolatedAverage`. Define the last-spin interpolation $\nu_{s,u}(F)$ from an explicit last-spin Gaussian decomposition, with its normalization, measurability, and endpoint identities.
12. `Cavity/Interpolation.lean:cavity_secondDerivative_bound`. Differentiate the interpolation twice, classify replica-edge terms using `Cavity/Coefficients`, and bound the remainder by $C_K(\text{thirdMoment} + N^{-3})$.
13. `FreeEnergy.lean:rs_freeEnergy_error`. Prove the product endpoint identity at $s = 0$, extend the open-interval derivative to the closed interval, and integrate `rsGap_deriv`. The completed `uniform_secondMoment` theorem then supplies the M/N upper bound.

6 Recommended proof roadmap

Model and Gaussian API. Add a finite-index Gaussian-law module. Decide whether the current smart-path structure remains law-based or gains an explicit affine realization. Finish obligations 1–2 and add endpoint lemmas for `fullPathHamiltonian`. This removes the common blocker for free-energy, pressure, and concentration proofs.

Scalar package. Finish obligations 3–5. Gaussian convolution and the scalar trial identity are complete. Prove the reference-measure normalization, then construct the PDE/diffusion order parameter and its uniform strict sign.

GT package. Finish obligations 6–8. The recursion should expose its multiplier derivatives and factorization theorem as named lemmas, since coercivity uses both.

Pressure and tails. Finish obligations 9–10. Reuse the same finite Gaussian concentration interface created for the smart path, rather than building a second probabilistic model.

Cavity package. Finish obligations 11–12 and connect the Taylor remainder to `cavityRemainder`. Prove a theorem that constructs `HasCavityRemainderBound` from the interpolation estimate. This connection is currently absent and is necessary to remove the hypothesis from a fully unconditional version of the main theorem.

Free-energy integration. Finish obligation 13. Absorption and its uniform second-moment estimate are already complete from the explicit cavity-remainder hypothesis. The existing `Replicon`, `CompactCorollary`, and `MainResult` proofs should then close by recompilation once their upstream obligations are filled.

Certification. Run `lakebuildLatalaMeetsAT`, search the source for `sorry`, `admit`, and project-specific `axiom`, and inspect `#printaxiomsSpinGlass.AT.quantitative_strictAT`. Success means that `sorryAx` is absent. Ordinary foundations such as choice and quotient soundness may remain.

7 Structural issues not counted among the 13 holes

- `Cavity/System.lean:cavity_system` is true by unfolding the definition of `cavityRemainder`. It does not itself formalize the cavity expansion. The analytic content is precisely the unconstructed bound described above.
- `MainResult.lean:quantitative_strictAT` assumes the existence of a constant satisfying `HasCavityRemainderBounddataC`. Thus removing all 13 occurrences is enough to remove the present explicit placeholders, but not enough to derive that hypothesis from the compact strict-AT assumptions.
- The free-energy comment saying that the external field is omitted is stale. The definition `pathFreeEnergy` now uses `fullPathHamiltonian`, which adds the deterministic field. What is still missing is a proved endpoint identity and the closed-interval calculus needed to integrate the derivative.
- `SmartPath.lean` specifies Gaussian marginals and their covariance, but not a concrete coordinate realization. This is adequate only after proving a finite-dimensional Gaussian-law uniqueness and concentration API. Otherwise the structure should be strengthened once, near the root of the import spine.